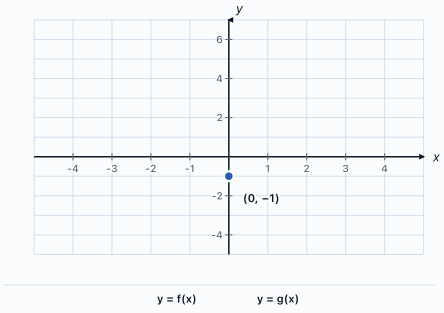
25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

SAAT Trigonometry, Conic Sections & 3D Vectors Practice Exercises

1 MCQ

The coordinate diagram shows the parabolic curve $f_1(x) = x^2 - 18x + 98$ in the domain of SAAT Trigonometry, Conics & Vectors. By completing the square $f_1(x) = (x-h)^2 + k$, find the sum $k + f_1(11)$.



A 47
B 21
C 38
D 36

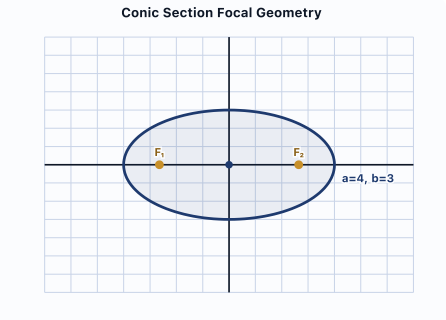
2 MCQ

A mathematical model in SAAT Trigonometry, Conics & Vectors is defined by the cubic function $y_2(x) = 2x^3 - 9x^2 + 15x - 14$. Calculate the exact gradient of the tangent to the curve at the point where $x = 3$.

A 69
B 0
C 15 + 1
D 15

3 MCQ

In SAAT Trigonometry, Conics & Vectors, two displacement vectors are given by $\mathbf{u}_3 = \langle 9, -8 \rangle$ and $\mathbf{v}_3 = \langle 9, 7 \rangle$. Compute the exact scalar value of $\mathbf{u}_3 \cdot \mathbf{v}_3 + \|\mathbf{u}_3\|^2$.



A 170
B 25
C 145
D 282

4 MCQ

The linear growth profile shown in the diagram for SAAT Trigonometry, Conics & Vectors follows an arithmetic progression with first term $a_1 = 10$ and common difference $d = 6$. Find the exact sum S_{15} of the first 15 terms.

A 94

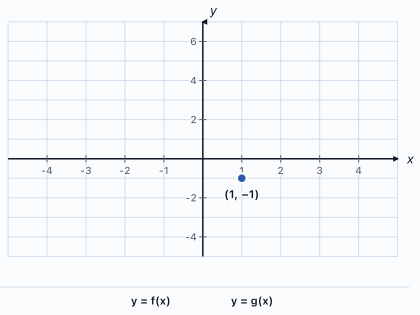
B 780

C 870

D 1410

5 MCQ

In SAAT Trigonometry, Conics & Vectors, evaluate the exact definite integral $I_5 = \int_0^2 (18x^2 + 36x)\,\mathrm{d}x$.



A 144

B 156

C 120

D 61

6 MCQ

In SAAT Trigonometry, Conics & Vectors, consider the quadratic function $f_6(x) = x^2 - 20x + 189$. Express $f_6(x)$ in the completed-square form $(x-h)^2 + k$ and compute the exact value of $k + f_6(17)$.

A 237

B 138

C 220

D 227

7 MCQ

The graph illustrates the cubic model $y_7(x) = 2x^3 - 10x^2 + 33x - 21$ studied in SAAT Trigonometry, Conics & Vectors. Find the instantaneous gradient $\frac{\mathrm{d}y}{\mathrm{d}x}$ of the tangent to the curve at $x = 8$.

A 257

B 417

C 224

D 61

8 MCQ

In SAAT Trigonometry, Conics & Vectors, two displacement vectors are given by $\mathbf{u}_8 = \langle 10, -7 \rangle$ and $\mathbf{v}_8 = \langle 8, 5 \rangle$. Compute the exact scalar value of $\mathbf{u}_8 \cdot \mathbf{v}_8 + \|\mathbf{u}_8\|^2$.

A 45

B 194

C 149

D 264

9 MCQ

For what value of k are the vectors $\mathbf{u} = \langle 2, k \rangle$ and $\mathbf{v} = \langle -3, 3 \rangle$ orthogonal (perpendicular)?

A -2

B 5

C 14

D 2

10 MCQ

For what value of k are the vectors $\mathbf{u} = \langle 3, k \rangle$ and $\mathbf{v} = \langle -8, 4 \rangle$ orthogonal (perpendicular)?

A -6

B 10

C 7

D 6

11 MCQ

For what value of k are the vectors $\mathbf{u} = \langle 4, k \rangle$ and $\mathbf{v} = \langle -15, 5 \rangle$ orthogonal (perpendicular)?

A 17

B 12

C 9

D -12

12 MCQ

For what value of k are the vectors $\mathbf{u} = \langle 5, k \rangle$ and $\mathbf{v} = \langle -24, 6 \rangle$ orthogonal (perpendicular)?

A -20

B 20

C 26

D 11

13 MCQ

For what value of k are the vectors $\mathbf{u} = \langle 6, k \rangle$ and $\mathbf{v} = \langle -35, 7 \rangle$ orthogonal (perpendicular)?

A 30

B 37

C -30

D 13

14 MCQ

For what value of k are the vectors $\mathbf{u} = \langle 7, k \rangle$ and $\mathbf{v} = \langle -2, 2 \rangle$ orthogonal (perpendicular)?

A 7

B -7

C 14

D 9

15 MCQ

For what value of k are the vectors $\mathbf{u} = \langle 8, k \rangle$ and $\mathbf{v} = \langle -6, 3 \rangle$ orthogonal (perpendicular)?

A 11
B -16
C 19
D 16

16 MCQ

For what value of k are the vectors $\mathbf{u} = \langle 2, k \rangle$ and $\mathbf{v} = \langle -12, 4 \rangle$ orthogonal (perpendicular)?

A 10
B 14
C -6
D 6

17 MCQ

For what value of k are the vectors $\mathbf{u} = \langle 3, k \rangle$ and $\mathbf{v} = \langle -20, 5 \rangle$ orthogonal (perpendicular)?

A 8
B 12
C 17
D -12

18 MCQ

For what value of k are the vectors $\mathbf{u} = \langle 4, k \rangle$ and $\mathbf{v} = \langle -30, 6 \rangle$ orthogonal (perpendicular)?

A -20
B 20
C 10
D 26

19 MCQ

For what value of k are the vectors $\mathbf{u} = \langle 5, k \rangle$ and $\mathbf{v} = \langle -7, 7 \rangle$ orthogonal (perpendicular)?

A 5
B 14
C -5
D 12

20 MCQ

For what value of k are the vectors $\mathbf{u} = \langle 6, k \rangle$ and $\mathbf{v} = \langle -4, 2 \rangle$ orthogonal (perpendicular)?

A 8
B 12
C 14
D -12

21 MCQ

For what value of k are the vectors $\mathbf{u} = \langle 7, k \rangle$ and $\mathbf{v} = \langle -9, 3 \rangle$ orthogonal (perpendicular)?

A 24

B 10

C 21

D -21

22 MCQ

For what value of k are the vectors $\mathbf{u} = \langle 8, k \rangle$ and $\mathbf{v} = \langle -16, 4 \rangle$ orthogonal (perpendicular)?

A 32

B -32

C 36

D 12

23 MCQ

For what value of k are the vectors $\mathbf{u} = \langle 2, k \rangle$ and $\mathbf{v} = \langle -25, 5 \rangle$ orthogonal (perpendicular)?

A -10

B 7

C 15

D 10

24 MCQ

For what value of k are the vectors $\mathbf{u} = \langle 3, k \rangle$ and $\mathbf{v} = \langle -6, 6 \rangle$ orthogonal (perpendicular)?

A 9

B -3

C 14

D 3

25 MCQ

For what value of k are the vectors $\mathbf{u} = \langle 4, k \rangle$ and $\mathbf{v} = \langle -14, 7 \rangle$ orthogonal (perpendicular)?

A -8

B 11

C 15

D 8